Beamsteering-Enabled Optical Wireless Integrated Sensing and Communication

Haitham AL Satai*, Bastien Béchadergue*, Luc Chassagne*, Kevin Acuna-Condori*,
Barthélemy Cagneau* and Pascal Ruaux*
*Laboratoire d'Ingénierie des Systèmes de Versailles
Université Paris-Saclay, UVSQ, 78140 Vélizy-Villacoublay, France
Email: {haitham.al-satai, bastien.bechadergue, luc.chassagne, kevin-jose.acuna-condori,
barthelemy.cagneau, pascal.ruaux}@uvsq.fr

***Abstract*—Integrated sensing and communication (ISAC) has been identified as a key technology for future wireless networks, and multiple solutions have therefore been proposed recently, including systems based on optical wireless communications (OWC). Although promising, these OWC-based ISAC systems often need several optical sources and/or photoreceivers, or even complementary sensors (cameras etc.), to perform positioning while ensuring good communication performance. Therefore, in this paper, we propose a new concept of beamsteering-enabled ISAC system that takes full advantage of the possibility, conferred by optical beamsteering, of modifying the orientation of a single source to enable both communication with an user equipment (UE) and two-dimensional positioning of this UE. The architecture of the proposed system is detailed and then validated through simulations, which show that the use of beamsteering enables both a substantial increase in communication coverage and simultaneous cm-level positioning.**

***Index Terms*—Integrated sensing and communication (ISAC), beamsteering, optical wireless communications (OWC), LiFi**

I. Introduction

In its vision document for the sixth generation (6G) of international mobile telecommunications (IMT), called *IMT-2030* [1], the International Telecommunications Union (ITU) identifies integrated sensing and communication (ISAC) as a key technology for next-generation IMT. ISAC consists in sharing the same resources in the time, frequency and space domains to enable both data transmission and sensing [2], especially positioning, with an expected sub-10 cm accuracy [1]. Such ISAC systems should eventually foster the emergence of new applications in massive Internet of Things, Industry 4.0, or even non-terrestrial networks [3].

Various technologies have been explored for ISAC, including optical wireless communications (OWC) [4]–[6]. In [5], a system based on phase difference of arrival for positioning and orthogonal frequency division multiplexing (OFDM) for data transmission was experimentally tested, achieving a data rate of 22 Mbps and a positioning error of less than 8 cm in 80% of cases on a 2×2 m^2 receiver plane. In [6], the authors of this article demonstrated performance of the same order – 12 Mbps link with positioning error under 5.9 cm in 90% of cases over 1.2×1.2 m^2 – by combining received signal strength (RSS) for positioning and multi-band carrierless amplitude and phase (m-CAP) modulation for communication.

Although OWC are beginning to demonstrate their potential for ISAC, these examples show that they suffer from significant limitations in terms of communication coverage. Several studies are therefore focusing on the use of optical beamsteering techniques to track and target user equipment (UE) by OWC access points (AP), and vice versa, allowing for a significant increase in coverage and, more generally, in quality of service [7]–[10]. In parallel, it should be noted that the OWC-based ISAC systems proposed so far require several optical sources and/or photoreceivers to perform positioning, which can be restrictive in some cases [11].

To address this issue, we have proposed and validated in [12] a new technique for positioning an UE in a two-dimensional plane, combining RSS and angle-of-arrival (AOA) measurements of direct-current optical (DCO) signals transmitted by a single optical source which orientation is modulated using beamsteering. However, this system could only perform positioning. Therefore, in this article, we propose a new beamsteering-enabled ISAC system that employs this positioning method, fed not by DCO signals but by DCO-OFDM signals to also support communication.

To do so, we consider a $4\times4\times2.5$ m^3 indoor room equipped with a single but steerable infrared (IR) light-emitting diode (LED) in the center of its ceiling, and serving an UE moving in a given plane. By successively orienting this source in three different directions and emitting each time a DCO-OFDM signal, it becomes possible to establish communication with the UE and estimate its position in this plane, according to methods that will be detailed in Section II. Given that the choice of these orientations, the source directivity or the orientation of the UE, may have an impact on ISAC performance, we have validated the operation of the proposed system by simulation and studied the influence of these parameters, as detailed in Section III. The results show that with a good directivity/orientations compromise, the UE position can be estimated with a single optical source at the cm level, with no loss of connectivity over almost the entire receiver plane (i.e. $\approx$ 16 m^2), at a data rate of over 50 Mbps, even with possible variations in UE orientation. The proposed ISAC system thus takes full advantage of beamsteering to optimize not only communication performance but also positioning, opening the way for promising future work, as concluded in Section IV.

II. ISAC System Architecture and Modeling

A. Overall System Architecture and Modeling

The beamsteering-based ISAC system we propose to study here consists, as shown in Fig. 1, of an AP equipped with one IR LED and placed at the center $\mathbf{T}$ of the ceiling of a room of dimensions $L \times W \times H$, and of an UE equipped with at least one photodiode (PD) and whose position $\mathbf{R}$ can vary within a receiving plane parallel to the floor and located at a distance $|h|$ from the ceiling. We note $[x, y, z]$ the coordinates of $\mathbf{R}$ in the main frame $\mathcal{F} = (\mathbf{T}, \mathcal{B})$ of origin $\mathbf{T}$ and orthonormal basis of $\mathbb{R}^3$ $\mathcal{B} = \{\mathbf{u}_x, \mathbf{u}_y, \mathbf{u}_z\}$.

It is assumed that both the AP and UE orientations, represented by the unit support vectors $\mathbf{n}_{t,i}$ and $\mathbf{n}_r$ respectively, can be modified in real time using any beamsteering system, the details of which are beyond the scope of this article. In addition, it is assumed that the AP is oriented in three different consecutive directions for the positioning method to work properly, as we will see in Section II-C. The mute index i in $\mathbf{n}_{t,i}$ therefore refers to the i-th orientation of the AP for a given position of the UE.

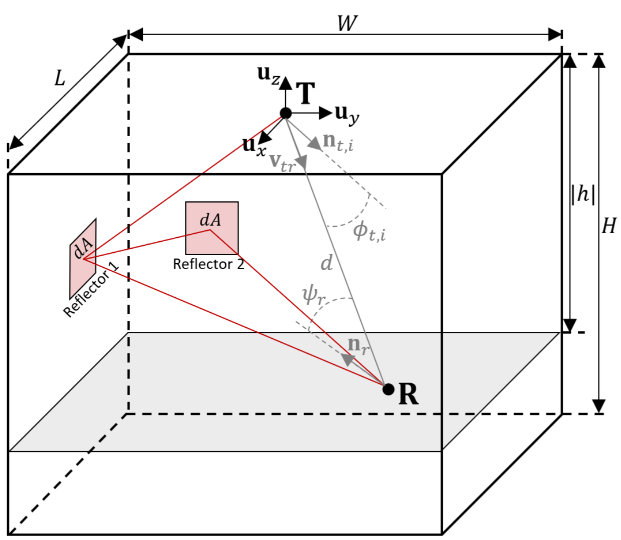

Fig. 1: Schematic view of the proposed beamsteering-based ISAC system and of the geometrical notations.

Based on such a geometrical configuration, the system operates as follows: the AP points to a given direction and emits a DCO-OFDM data signal of instantaneous optical power $P_{t,i}(t)$, which then propagates to the UE via a direct line-of-sight (LOS) path and a multitude of indirect non-LOS (NLOS) paths. It is assumed here that the signal bandwidth is less than the coherence bandwidth of the OWC channel, so that the delays introduced by the various NLOS paths are negligible. We can therefore group together under a single H_{NLOS} term the gains of these NLOS paths, which alongside the gain H_{LOS} of the direct LOS path means that the signal received by the UE $y_i(t)$ will be of the form:

$$y_i(t) = R_{\mathrm{PD}} P_{t,i}(t) \left(H_{\mathrm{LOS}} + H_{\mathrm{NLOS}}\right) + n(t), \tag{1}$$

where R_{PD} is the PD sensitivity and $n(t)$ is the additive white Gaussian noise (AWGN) at the PD level.

In order to model the signal received by the UE in more detail, it is necessary to clarify the expressions for H_{LOS} and H_{NLOS}, which can be done by first assuming that the LED source used by the AP has a Lambertian pattern of emission of order m_t, where m_t depends on the semi-angle at half power $\Phi_{1/2}$ of the LED, i.e. on its directivity, according to the relation $m_t = -\ln 2 / \ln\left(\cos\left(\Phi_{1/2}\right)\right)$. We then know that in such a case, the LOS gain is given by [13]:

$$H_{\mathrm{LOS}} = \begin{cases} \frac{A_{\mathrm{PD}}(m_t+1)}{2\pi d^2} \cos^{m_t}(\phi_{t,i}) \cos(\psi_r) \\ \text{if } 0 \le \phi \le \pi/2 \text{ and } 0 \le \psi_r \le \Psi_{\mathrm{c}}, \\ 0 \text{ otherwise,} \end{cases} \tag{2}$$

where A_{PD} is the sensitive area of the PD, $\phi_{t,i}$ is the angle of emission, i.e. the angle between the AP's normal $\mathbf{n}_{t,i}$ and the direction $\mathbf{v}_{tr}$ of the LOS link between AP and UE, and ψ_r is the angle of reception, i.e. the angle between the UE's normal $\mathbf{n}_r$ and the direction $-\mathbf{v}_{tr}$ of the LOS link between UE and AP. Note that according to (2), the transmitted signal can only be collected by the PD if its angle of incidence ψ_r is smaller than the field of view (FoV) Ψ_c of the UE.

On its side, the NLOS gain can be evaluated using the frequency-domain method proposed by Schulze in [14]. This consists in dividing all the walls in the environment into a set of N_w reflectors, e.g. 'Reflector 1' in Fig. 1, which then behave as passive Lambertian sources of order 1 with a specific reflectivity coefficient. Using relation (2), we can evaluate: the gains of the direct paths from the AP to each reflector, then gathered in a matrix $\mathbf{t}$ of size $N_w \times 1$; the direct gains from one reflector to another, gathered in an $N_w \times N_w$ matrix $\mathbf{H}$; and the direct gains from each reflector to the UE, gathered in an $N_w \times 1$ matrix $\mathbf{r}$. In [14], it is then shown that the total NLOS gain, H_{NLOS}, taking into account reflections up to infinite orders can be obtained using:

$$H_{\mathrm{NLOS}} = \mathbf{r}^T \mathbf{G}_\rho \left(\mathbf{I} - \mathbf{H}\mathbf{G}_\rho\right)^{-1} \mathbf{t}, \tag{3}$$

where $\mathbf{I}$ is the identity matrix of dimension $N_w \times N_w$ and $\mathbf{G}_\rho$ is the $N_w \times N_w$ diagonal matrix which elements are the reflectivity coefficients of each reflector.

Once received, the signal $y_i(t)$ is processed by the UE to retrieve the transmitted data on the one hand, as further detailed in Section II-B, and to estimate its location on the other hand, as explained in Section II-C.

B. How Communication Works

The communication function of our ISAC system is based on the transmission, from the AP to the UE, of a data signal modulated using a common DCO-OFDM scheme [15], which consists in dividing the available bandwidth B between N_{sc} orthogonal subcarriers, each loaded with binary data via quadrature amplitude modulation (QAM) symbols of order M. The specificity of DCO-OFDM lies in the fact that only the first half of the subcarriers is loaded with M-QAM symbols, the second half being then obtained by Hermitian symmetry. Null subcarriers are also added at the center and at one end of the spectrum to avoid aliasing, and a cyclic prefix of length $r_{cp}N_{sc}$ is then added in the time domain to fight inter-symbol interference and enable equalization.

The consecutive DCO-OFDM symbols thus eventually produced form the electrical data signal $x_i(t)$ that is transmitted in its optical form $P_{t,i}(t)$ by the IR LED of the AP. Note here that to avoid photo-biological issues [16], the average value of $P_{t,i}(t)$ must remain below a maximum optical power $P_{t,max}(\Phi_{1/2})$, which depends on the source directivity according to [9]:

$$P_{t,max}(\Phi_{1/2}) = \frac{8\pi \ln\left(\cos\left(\Phi_{1/2}\right)\right)}{\ln\left(\cos\left(\Phi_{1/2}\right)\right) - \ln 2}. \tag{4}$$

After LOS and NLOS propagation, the transmitted optical signal is collected by the UE and turned into the received electrical signal $y_i(t)$, which is re-synchronized, processed, equalized and de-mapped to recover the transmitted data, at a final data rate R_b given by:

$$R_b = \frac{(N_{sc}-2)B\log_2(M)}{2N_{sc}(1+r_{cp})}. \tag{5}$$

C. How Positioning Works

In addition to the data it contains, the signal $y_i(t)$ received by the UE for a certain orientation i of the AP can be exploited to obtain an estimate $\hat{P}_{r,i}(t)$ of the optical power $P_{r,i}(t)$ incident to the UE. From (1), we deduce that simply dividing $y_i(t)$ by the photosensitivity of the PD R_{PD} gives:

$$\hat{P}_{r,i}(t) = P_{r,i}(t) + \frac{n(t)}{R_{\mathrm{PD}}}, \tag{6}$$

with $P_{r,i}(t) = P_{t,i}(t)\left(H_{\mathrm{LOS}} + H_{\mathrm{NLOS}}\right)$. We can therefore observe that this estimate contains an error due to the noise at the receiver level.

Let us now assume that for the same position of the UE, the AP is then oriented in a second direction j, so that an estimate $\hat{P}_{r,j}(t)$ of the incident optical power can be obtained. Noting $\hat{P}_{r,i}$ and $\hat{P}_{r,j}$ the average values of these two signals and assuming the noise and NLOS components are negligible, we can show that:

$$\frac{\hat{P}_{r,i}}{\hat{P}_{r,j}} = \left(\frac{\cos\phi_{t,i}}{\cos\phi_{t,j}}\right)^{m_t}, \tag{7}$$

which is remarkable in that this ratio depends only on the orientation of the AP and the position of the UE, through the variables $\phi_{t,i}$ and $\phi_{t,j}$, but neither on the transmitted optical power nor on the orientation of the receiver.

Adopting the notations $K_{ij} = \left(\frac{\hat{P}_{r,i}}{\hat{P}_{r,j}}\right)^{\frac{1}{m_t}}$ and $\mathbf{n}_{t,i} = [a_i, b_i, c_i]$, and then noting that $\cos\phi_{t,i} = \mathbf{v}_{tr} \cdot \mathbf{n}_{t,i}$, with '$\cdot$' the scalar product operator, we can eventually show that if the z coordinate of the UE is known (e.g. $z = h$ here), then its x and y coordinates can be deduced using [12]:

$$x = \frac{(K_{ik}b_k - b_i)(c_i - K_{ij}c_j) - (K_{ij}b_j - b_i)(c_i - K_{ik}c_k)}{(K_{ik}b_k - b_i)(K_{ij}a_j - a_i) - (K_{ij}b_j - b_i)(K_{ik}a_k - a_i)} h, \tag{8}$$

$$y = \frac{(K_{ik}a_k - a_i)(c_i - K_{ij}c_j) - (K_{ij}a_j - a_i)(c_i - K_{ik}c_k)}{(K_{ik}a_k - a_i)(K_{ij}b_j - b_i) - (K_{ij}a_j - a_i)(K_{ik}b_k - b_i)} h, \tag{9}$$

where i, j and k are indices referring to three different orientations of the AP. In other words, for a given position of the UE, by transmitting a signal with an AP having three different consecutive orientations, it is possible for the UE to estimate its position in a reception plane of known height, regardless its orientation and the transmitted optical power. Then, as long as the consecutive AP orientations do not induce a link loss (e.g. if one of these orientations is opposite to the UE's direction), data transmission from the AP to the UE is possible along with UE positioning, so that the proposed beamsteering-enabled ISAC system may operate properly.

III. Simulation Validation and Discussion

A. Simulation Parameters

To validate and evaluate our system, we carried out simulations with MATLAB in the environment shown in Fig. 1, with the parameters listed in Table I. The walls of a $4\times$ 4×2.5 m^3 room are thus divided into reflectors of area dA = 20×10 cm^2 (as smaller values led to excessive simulation times), and of common reflectivity ρ = 0.6 [17]. An IR LED of modulation bandwidth B = 30 MHz and variable semi-angle at half power $\Phi_{1/2}$ is then placed in the center of the ceiling, and emits a DCO-OFDM signal whose average optical power matches the photobiological safety limit defined in (4) for each $\Phi_{1/2}$.

This signal enables communication with the UE by dividing the available bandwidth B into N_{sc} = 32 subcarriers, of which N_d = 15 are used for data transmission. Each data subcarrier is then modulated using a QAM modulation of order M varying from 2 to 256, so that data rates R_b ranging from 14 Mbps to 112.5 Mbps may be reached. In practice, around 35,000 M-QAM symbols are transmitted for each position of the UE and orientation of the AP, so that the bit error rate (BER) quantum is of the order of 10^{-5}, well below the 3.8×10^{-3} threshold used to define the communication coverage area [18].

In parallel, localization of the UE is ensured by successively orienting the LED according to the three vectors $\mathbf{n}_{t,i/j/k}$ defined in Table I, which point to the real coordinates of the UE ($[x, y, h]$, with h known) with a misalignment of $\pm r_t$ on the $\mathbf{u}_x$ and $\mathbf{u}_y$ axes. Whatever the actual position of the UE, the AP points therefore successively to three points on a circle of radius r_t around the UE. This 'pointing radius' r_t is considered to vary from 0 cm (perfect pointing, no positioning possible) to 100 cm (coarse pointing) in order to study its impact on ISAC performance. Our system is thus assumed to already have coarse knowledge of the UE's location, which can be done using the approach described in [12], where the UE's position is detected and refined iteratively.

Finally, the UE always points to the AP (except in Section III.B.2) and its PD has a sensitive area of 26.4 mm^2, i.e. the area of a single Hamamatsu S6967 PD [19], compared with 105.6 mm² (i.e. four S6967) in our previous work [9], so differences in performance are expected. In addition, the electronic noise of the receiver is considered to have a power spectral density (PSD) of 10^{-21} W/Hz, which leads to a signal-to-noise ratio (SNR) of at least 23 dB over 90% of the receiver plane, a common value in indoor OWC [20].

TABLE I: SIMULATION PARAMETERS AND VALUES.

Room parameters	
Parameter	**Value**
Dimensions ($L \times W \times H$)	4×4×2.5 m^3
Area of the wall reflectors (dA)	20×10 cm^2
Walls reflectivity (ρ)	0.6 [17]
AP parameters	
Parameter	**Value**
Coordinates	$\mathbf{T} = [0, 0, 0]^T$
Semi-angle at half-power ($\Phi_{1/2}$)	From 5° to 60°
Transmitted optical power (P_t)	$P_{t,max}(\Phi_{1/2})$ (4)
Source orientations ($\mathbf{n}_{t,i/j/k}$)	$\mathbf{n}_{t,i} = [x \pm r_t, y, h]^T / \lvert\lvert \mathbf{n}_{t,i} \rvert\rvert$ $\mathbf{n}_{t,j} = [x, y - r_t, h]^T / \lvert\lvert \mathbf{n}_{t,j} \rvert\rvert$ $\mathbf{n}_{t,k} = [x, y + r_t, h]^T / \lvert\lvert \mathbf{n}_{t,k} \rvert\rvert$
Pointing radius (r_t)	From 0 cm to 100 cm
DCO-OFDM parameters	
Parameter	**Value**
Signal bandwidth (B)	30 MHz
Total number of sub-carriers (N_{sc})	32
Number of data sub-carriers (N_d)	15
QAM order (M)	From 2 to 256
Overall data rate (R_b)	From 14 Mbps to 112.5 Mbps
UE parameters	
Parameter	**Value**
Coordinates	$\mathbf{R} = [x, y, h]^T$
Height of the UE ($H - \lvert h \rvert$)	0.85 m (i.e. $\lvert h \rvert$ = 1.65 m)
FOV of the PD (Ψ_c)	85°
Responsivity of the PD (R_{PD})	0.63 A/W
Effective area of the PD (A_{PD})	26.4 mm^2
Orientation of the PD ($\mathbf{n}_r$)	Toward $\mathbf{T}$
Noise PSD (N_0)	10^{-21} W/Hz

B. Positioning Performance

First, we evaluated the positioning performance of our ISAC system, by moving the UE in 10 cm steps along the $\mathbf{u}_x$ and $\mathbf{u}_y$ axes of the reception plane (i.e. 1681 positions in total). For each position and then each orientation of the AP, we simulated the transmission of 35,000 M-QAM symbols modulated in DCO-OFDM. The powers of the three received signals were then used to estimate with (8) and (9). For a given set of parameters, 1681 estimates were thus obtained, and then compared with the actual positions of the UE to determine their root mean square (RMS) error. It was eventually possible to determine, from the empirical cumulative distribution function (CDF) of these 1681 RMS errors, the value under which 90% of them are, noted $\varepsilon_{90\%}$ and used as the main positioning performance metric here.

1) Performance in an ideal case: Figure 2 shows the evolution of the RMS error $\varepsilon_{90\%}$ with the pointing radius r_t and the IR LED directivity $\Phi_{1/2}$, in the case where the UE points perfectly towards the AP. First, we can clearly see that for low values of r_t, the most directive sources provide the best positioning performance. Thus, with $\Phi_{1/2}$ = 5°, the minimum positioning error is 0.26 cm, whereas this error reaches at best 26.49 cm with a 45° source (and 57.78 cm with a 60° source, not shown in Fig. 2).

This difference is due to the NLOS gains, which are much greater when the source is diffuse. Since we remain right under the photobiological safety limit, greater optical power is indeed sent in this case (e.g. 8.37 W at 45° against 0.13 W at 5°), over a wider range of directions, thus increasing the probability of receiving powerful indirect signals. Therefore, in the case where N_0 = 0 W/Hz (no noise) and r_t = 0.1 m, $\varepsilon_{90\%}$ is infinitesimal ($< 10^{-13}$) whatever the source directivity when only the LOS gain is taken into account, but increases when the NLOS gain is also included, reaching 26.41 cm when $\Phi_{1/2}$ = 45° versus only 0.26 cm when $\Phi_{1/2}$ = 5°.

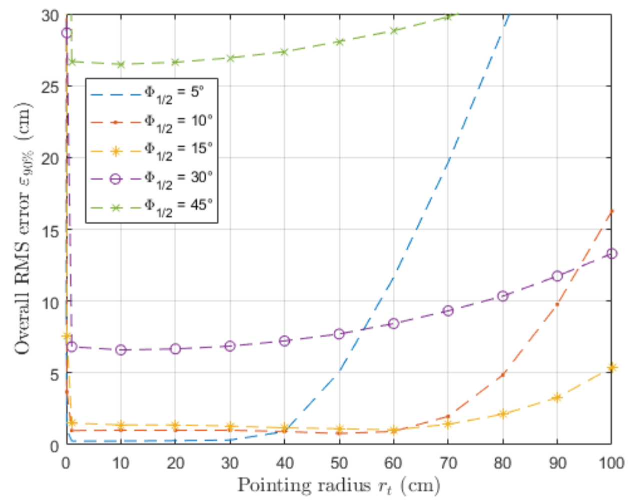

Fig. 2: Evolution of overall RMS error $\varepsilon_{90\%}$ with the pointing radius r_t and the semi-angles at half power $\Phi_{1/2}$.

In parallel, we observe that whatever the source directivity, $\varepsilon_{90\%}$ increases with r_t, i.e. as the AP is oriented toward points further away from the actual position of the UE. Increasing r_t leads indeed to SNR degradation, which converts into a larger positioning error, a phenomenon that becomes even more pronounced as the source becomes more directive, as shown in Fig. 2 when $\Phi_{1/2}$ = 5°, where the error soars up when $r_t > 40$ cm. In such cases, the Lambertian pattern is such that the power emitted for emission angles $\phi_{t,i}$ greater than $\Phi_{1/2}$ decreases rapidly with $\phi_{t,i}$ and therefore with r_t, leading to an even faster degradation of the SNR and eventually of $\varepsilon_{90\%}$. Finally, note that the positioning error for very low values of r_t is very large, which logically stems from the very small variation in emission pattern for very low emission angles $\phi_{t,i}$. The three orientations are therefore too difficult to distinguish from each other for accurate positioning.

2) Sensitivity to receiver's misalignment: An analysis of the error induced by inaccurate AP orientations is proposed in [12] and therefore not reproduced here. We may however wonder whether variations in receiver orientation between two position measurements, or even during the same measurement, can have an impact on the positioning error. Therefore, we now consider the case where the UE points to the AP with an error, and more precisely the case where the straight line directed along the UE's normal vector $\mathbf{n}_r$ crosses the ceiling of the room at the point $[\delta x, \delta y, 0]$ and not $[0, 0, 0]$, with δx and δy the coordinates of the pointing error modeled by uniform random variables between -δ_r and δ_r.

First, it is assumed that δx and δy do not change as the source is successively oriented in the three directions required to estimate the UE's position, and that the pointing radius r_t is set to 10 cm. In such a case, we observe that for a directive source of $\Phi_{1/2}$ = 5°, the overall RMS error $\varepsilon_{90\%}$ remains identical even if δ_r = 1 m, i.e. if the receiver's pointing error potentially rises up to ± 1 m along the $\mathbf{u}_x$ and/or $\mathbf{u}_y$ axes. For a more diffuse source of $\Phi_{1/2}$ = 45°, the variation in $\varepsilon_{90\%}$ when δ_r ranges from 0 to 1 m is 1.7 cm only. In other words, and in line with the conclusions of Section II-C, our positioning method does not depend on the UE's orientation, as long as the latter does not change throughout the positioning process and with sufficiently directive sources. In the case of more diffuse sources, a relatively small additional error arises due to the variations in collected optical power induced by more powerful NLOS signals.

In practice, however, the UE's pointing error is likely to vary from one orientation of the AP to the next. From a theoretical point of view, such variations should lead to positioning errors as the reception angle ψ_r will now change between orientations i, j and k, which means that the $\cos(\psi_r)$ terms in (2) will no longer cancel each other out when calculating K_{ij} with (7). Figure 3 shows in such a case the variation in $\varepsilon_{90\%}$ as a function of δ_r. We can see that, overall, the greater δ_r, the greater $\varepsilon_{90\%}$, whatever the source directivity. However, the most directive sources are much more robust against increases in δ_r, which reflects again the lesser importance of NLOS paths in this case.

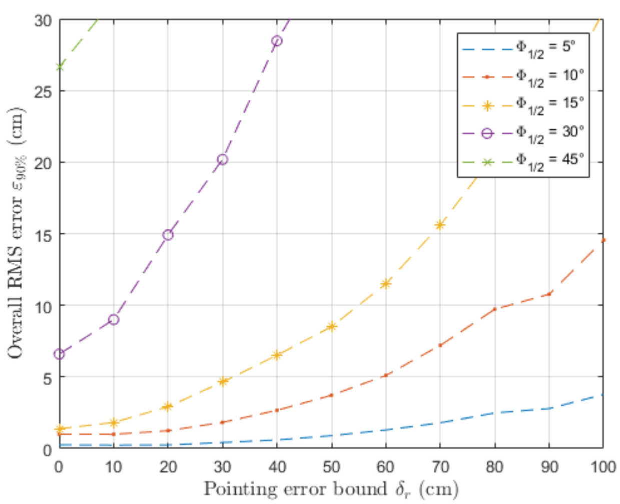

Fig. 3: Evolution of the error $\varepsilon_{90\%}$ with the receiver pointing error bound δ_r and the semi-angles at half power $\Phi_{1/2}$.

All these results mean, in conclusion, that the proposed ISAC system enables positioning, with an error of the order of a cm when the source used is fairly directive and the AP's orientations are carefully chosen, despite possible pointing errors from the UE to the AP. Note also that we did not observe any variations in positioning performance with the QAM order M, as the average power of the received signal does not change with this parameter.

C. Communication Performance

The communication performance of our ISAC system was then evaluated following the same principles as in Section III-B, this time focusing on the communication coverage metric, defined as the area of the receiver plane over which BER $< 3.8 \times 10^{-3}$ [18]. More precisely, for a given position of the UE, we consider the latter to be in the coverage zone of the AP if the BER of the three links established with the three successive orientations of the IR LED required for position estimation are all less than 3.8×10^{-3}.

1) Case of a fixed QAM order: First, we consider the case of a fixed QAM order M = 16. Figure 4 shows in this case the evolution of the communication coverage as a function of r_t and $\Phi_{1/2}$. Overall, we can observe that the coverage decreases more or less rapidly with r_t, as a consequence of SNR degradation induced by LED pointing further away from the UE. However, this degradation is less pronounced for the most diffuse sources (30 and 45°) which emit higher powers over a wider range of directions, and therefore benefit from higher NLOS paths. This phenomenon also explains why the coverage remains larger for diffuse sources compared to directive sources, despite perfect pointing (r_t = 0 cm).

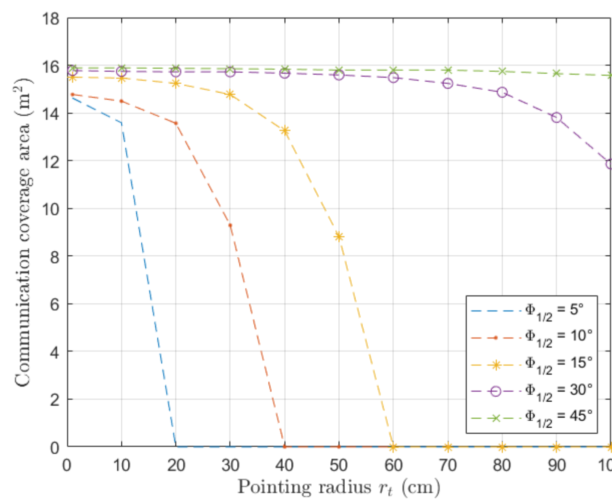

Fig. 4: Evolution of the coverage with the pointing radius r_t and the semi-angle at half power $\Phi_{1/2}$ (M = 16).

2) Incidence of the QAM order: Similar observations can be made about Fig. 5, which shows the evolution of communication coverage as a function of QAM order M and of the semi-angle at half-power $\Phi_{1/2}$, when r_t = 0.1 cm, i.e. for a set of orientations that allows for a good compromise between positioning error and coverage. It can be seen that, whatever $\Phi_{1/2}$, the coverage decreases with M, which can be explained by the increase with M of the SNR required to maintain a target BER. At the same time, we can see that the most directive sources have lower coverage performance than the most diffuse sources, regardless of the QAM order, which again stems from the greater contribution of NLOS paths to the received signal, i.e. to the SNR in the former case.

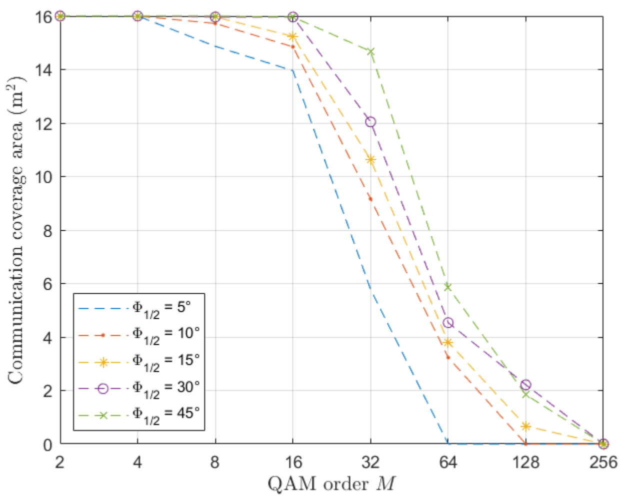

Fig. 5: Evolution of the coverage with the QAM order M and the semi-angle at half power $\Phi_{1/2}$ (r_t = 10 cm).

We can conclude from these results that the proposed ISAC system supports communication links over almost the entire room, at least for QAM orders ranging from 2 to 16, i.e. for data rates between 14 Mbps and 56.25 Mbps. For comparison, in a similar configuration with $M = 16$ but without beamsteering (i.e. $\mathbf{n}_{t,i/j/k} = [0, 0, -1]$ and $\mathbf{n}_r = [0, 0, 1]$), the coverage is 6.30 m² with $\Phi_{1/2} = 45°$ and only 0.11 m² with $\Phi_{1/2} = 5°$. Beamsteering thus increases coverage by a factor of at least 2.5 and up to 125, with the added advantage of precise positioning in the latter case. Note that in terms of data rate, the performance reached here could be improved, even for higher QAM orders, by increasing the sensitive area of the UE and using modulations with a lower peak-to-average power ratio, as we did in our previous work [9]. The signal bandwidth could also be extended using faster sources like lasers, in which case the positioning method would still work by replacing the Lambertian pattern by a Gaussian one.

IV. Conclusions and Future Works

In this paper, we propose a new beamsteering-enabled ISAC system that exploits the possibility of modifying the orientation of an AP's optical source to communicate with an UE and estimate its position. By deliberately introducing three successive and different misalignment of the AP towards the UE, our method enables fine estimation of its position in a receiver plane of known height. Using simulations in an indoor OWC environment and taking into account direct and indirect optical propagation paths, we have shown that this system performs particularly well with directional sources (e.g. with $\Phi_{1/2} = 5°$ to $15°$) operating just below the photobiological-safety limit, providing both good communication coverage (at least 85% of the room) and UE positioning at the cm level, while remaining robust to possible misalignment of the latter with the transmitter, even during the positioning process itself. However, these purely theoretical results still need to be validated experimentally, a task we are currently undertaking, coupling the position estimation method described here with machine learning techniques to improve its performance.

Acknowledgment

This work was supported in part by the French National Research Agency (ANR) through the Project SAFELiFi under Grant ANR-21-CE25-0001-01, in part by the French Agence de l'innovation de défense (AID) through the project SOLERIS under Grant 2022 29 0951, and in part by the European Union's Smart Networks and Services Joint Undertaking (SNS JU) under grant agreement No. 101139292.

References

[1] *Framework and overall objectives of the future development of IMT for 2030 and beyond*, International Telecommunications Union, Recommendation ITU-R M.2160-0, Nov. 2023.

[2] A. Liu *et al.*, "A survey on fundamental limits of integrated sensing and communication," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 2, pp. 994–1034, 2022.

[3] D. K. Pin Tan *et al.*, "Integrated sensing and communication in 6G: Motivations, use cases, requirements, challenges and future directions," in *1st IEEE Int. Symp. Joint Commun. Sens. (JC&S)*, 2021, pp. 1–6.

[4] S. M. Kouhini *et al.*, "LiFi positioning for Industry 4.0," *IEEE J. Sel. Topics in Quantum Electron.*, vol. 27, no. 6, pp. 1–15, 2021.

[5] H. Yang *et al.*, "An advanced integrated visible light communication and localization system," *IEEE Trans. Commun.*, vol. 71, no. 12, pp. 7149-7162, 2023.

[6] L. Shi *et al.*, "Experimental Demonstration of Integrated Optical Wireless Sensing and Communication," *J. Lightw. Technol.*, vol. 42, no. 20, pp. 7070-7084, 2024.

[7] T. Koonen *et al.*, "High-Capacity Optical Wireless Communication Using Two-Dimensional IR Beam Steering," *J. Lightw. Technol.*, vol. 36, no. 19, pp. 4486-4493, Oct. 2018.

[8] R. Singh *et al.*, "Design and Characterisation of Terabit/s Capable Compact Localisation and Beam-Steering Terminals for Fiber-Wireless-Fiber Links," *J. Lightw. Technol.*, vol. 38, no. 24, pp. 6817–6826, Dec. 2020.

[9] H. A. Satai *et al.*, "Coverage Optimization With Beamsteering-Based Indoor Optical Wireless Communications," in *IEEE Int. Conf. Commun. (ICC)*, 2023, pp. 1149-1154.

[10] A. R. Ndjiongue *et al.*, "Double-Sided Beamforming in VLC Systems Using Omni-Digital Reconfigurable Intelligent Surfaces," *IEEE Commun. Mag.*, vol. 62, no. 2, pp. 150-155, 2024.

[11] Z. Zhu *et al.*, "A Survey on Indoor Visible Light Positioning Systems: Fundamentals, Applications, and Challenges," *IEEE Commun. Surveys Tuts.*, Early access, 2024.

[12] L. Chassagne *et al.*, "Optical Wireless Positioning by Modulation of the Optical Source Orientation," *Optics Express*, vol. 33, no. 6, pp. 13161-13183, 2025.

[13] Z. Ghassemlooy, L. N. Alves, S. Zvanovec, and M. A. Khalighi, Eds., *Visible Light Communications: Theory and Applications*. Boca Raton, FL, USA: CRC press, 2017.

[14] H. Schulze, "Frequency-Domain Simulation of the Indoor Wireless Optical Communication Channel," *IEEE Trans. Commun.*, vol. 64, no. 6, pp. 2551-2562, 2016.

[15] X. Zhang *et al.*, "The Evolution of Optical OFDM," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1430-1457, 2021.

[16] *Photobiological safety of lamps and lamp systems*, IEC-62471:2006, International Electrotechnical Commission, Geneva, Switzerland, 2006.

[17] P. Combeau *et al.*, "Material characterisation for optical wireless communication combining measurement and Monte Carlo simulations," *IET Optoelectronics*, vol. 17, no. 4, pp. 149–161, 2023.

[18] *Forward Error Correction for High Bit-Rate DWDM Submarine Systems*, Recommendations ITU-T G.975.1, International Telecommunication Union, Geneva, Switzerland, Feb. 2004.

[19] *Si PIN Photodiodes S2506/S6775/S6967 Series*, Hamamatsu, Bridgewater, NJ, USA, 2021.

[20] Y. Hei *et al.*, "Energy-Spectral Efficiency Tradeoff in DCO-OFDM Visible Light Communication System," *IEEE Trans. Veh. Technol.*, vol. 68, no. 10, pp. 9872–9882, Oct. 2019.